Tutorial 6

1. Suppose X is a continuous random variable. Write the CDF and PDF of

$$Y = \frac{1}{X}$$

in terms of the CDF and PDF of X.

2. If X is an exponential random variable with parameter $\lambda = 1$, compute the probability density function of the random variable $Y = \log X$.

3. If X is uniformly distributed over $(0, 1)$, find the density function of $Y = e^X$.

4. If X is uniformly distributed over $(-1, 1)$, find

   (I) $P\{|X| > \frac{1}{2}\}$.

   (II) The density function of the random variable $|X|$.

5. The Opportunity rover is trying to estimate the distance it has traveled, denoted x, since it landed. By matching its surroundings with an onboard map, it has a noisy measure of this distance. The measurement follows a Gaussian distribution, written as $X \sim \mathcal{N}(\mu, \sigma^2)$.

   (I) The rover has stopped driving before sundown. Its current estimate of distance traveled is $\mu = 500\,\text{m}$ and $\sigma^2 = 2.25\,\text{m}^2$. What is the probability that it has actually driven less than 500 m?

   (II) What is the probability that it has driven exactly 501 m?

   (III) What is the probability that it has actually driven more than 501 m? More than 505 m?

6. Let X be a continuous random variable with probability density function

$$f_X(x) = \begin{cases} \dfrac{C}{1 + (x + K)^2}, & -K - 1 \le x \le -K + 1, \\ 0, & \text{otherwise}, \end{cases}$$

where $C > 0$ and K are constants.

   (I) Determine C and K such that $E[X] = 5$.

   (II) Calculate $\text{Var}(X)$.

7. Spirit has approached a large crater on Mars and would like to send back an image for scientists on Earth to analyze. The image is made up of a sequence of pixel values transmitted over its low-gain antenna, which introduces errors with probability $p = 0.03$.

   (I) What is the probability mass function of N, the number of error-free pixels between erroneous pixels?

   (II) What is $E[N]$?

   (III) An upgraded rover, Curiosity, is being designed as the successor to Opportunity and Spirit. One of its requirements is a 99% probability that at least 1000 pixels in an image are received correctly before a bad one occurs. What should its value of p be?